

Introduction to Cybersecurity

Cybersecurity refers to the practice of protecting computer systems, networks, and data from unauthorized access, damage, or theft. As organizations and individuals increasingly rely on digital systems for everyday operations, the importance of robust cybersecurity measures has grown substantially. The field encompasses a wide range of practices, including network security, which focuses on protecting the infrastructure that connects computers and devices, and application security, which addresses vulnerabilities within software programs. Information security more broadly deals with protecting the confidentiality, integrity, and availability of data, a principle often referred to as the CIA triad. Cyber threats come in many forms, including malware, which is malicious software designed to damage or gain unauthorized access to systems, and phishing, a social engineering technique where attackers trick individuals into revealing sensitive information such as passwords or financial details. Ransomware, a particularly damaging type of malware, encrypts a victim's files and demands payment in exchange for the decryption key, and has caused significant financial and operational damage to organizations worldwide in recent years. Defending against these threats requires a layered approach, often referred to as defense in depth, which combines firewalls, antivirus software, intrusion detection systems, employee training, and regular security audits. As cyber threats continue to evolve in sophistication, cybersecurity professionals must stay current with emerging attack techniques and continuously update their defensive strategies to protect critical systems and sensitive information.